Training Day 7

1. Topics Covered and Brief Explanation

1. Introduction to Pandas

Pandas is a Python library used for data manipulation and analysis. It provides easy-to-use data structures that help in organizing, cleaning and analysing datasets. It is widely used in Data Science and Machine Learning.

2. Features of Pandas

- Handles structured data efficiently.
- Supports CSV and Excel file formats.
- Provides functions for filtering and sorting data.
- Helps in handling missing values.
- Supports data analysis and manipulation.

3. Pandas Series

A Series is a one-dimensional labelled data structure that can store values of different data types.

Example:

```
import pandas as pd

marks = pd.Series([85, 90, 78, 95])

print(marks)
```

4. Pandas DataFrame

A DataFrame is a two-dimensional data structure consisting of rows and columns. It is commonly used to represent tabular datasets.

Example:

```
import pandas as pd

data = {

   "Name": ["Aman", "Riya", "Karan"],

   "Marks": [85, 90, 78],

   "Age": [20, 21, 22]

}

df = pd.DataFrame(data)

print(df)
```

5. Reading and Displaying CSV Files

Learned how to import datasets stored in CSV format using the read_csv() function and display their contents using Pandas.

Example:

```
import pandas as pd

df = pd.read_csv("students.csv")

print(df.head())

print(df.info())
```

6. Basic Data Manipulation

Studied common operations used to explore and modify datasets.

- head() – Displays the first five rows.
- tail() – Displays the last five rows.
- info() – Provides information about columns and data types.
- describe() – Displays statistical information.
- sort_values() – Sorts data based on a column.

2. Tools and Technologies Used

Tool	**Purpose**
Python 3	Writing Python programs.
Pandas	Data manipulation and analysis.
NumPy	Supporting numerical operations.
Google Colab	Executing Pandas programs.
CSV Dataset	Practising data handling operations.

3. Practical Learning / Activities

- Created Pandas Series and DataFrames.
- Practised converting dictionary data into DataFrames.
- Loaded sample CSV datasets using read_csv().
- Displayed and explored dataset information.

4. Learning Outcomes

- Understood the purpose and features of Pandas.
- Learned the difference between Series and DataFrames.
- Gained knowledge of importing CSV datasets.